

Employee Attrition Prediction using Machine Learning

Internship Project • Week 2 Report | Kunal Nigewan

Objective

The objective of this project was to analyze employee attrition patterns and identify the key factors influencing employees leaving the company. The project focused on understanding employee behavior, workplace satisfaction, and retention-related trends using data analysis and machine learning techniques.

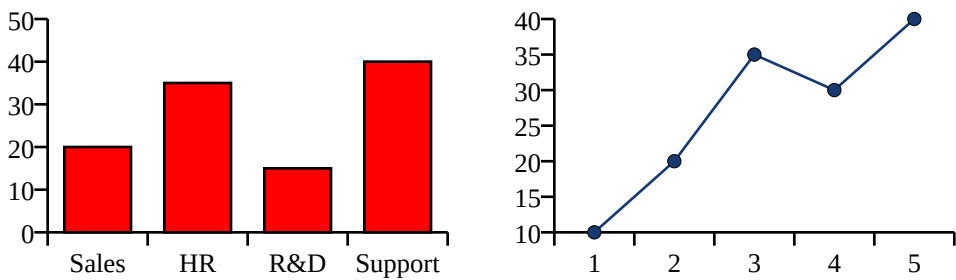
Procedure Loaded and explored the HR Attrition dataset using Pandas. Checked for missing values and verified data quality. Removed unnecessary columns such as EmployeeNumber, Over18, and StandardHours. Converted categorical columns into numerical format using One-Hot Encoding. Scaled numerical features using StandardScaler. Performed exploratory data analysis with multiple visualizations. Trained Logistic Regression, Random Forest, and Gradient Boosting models. Compared models using accuracy, recall, precision, and F1-score.

Model Results

Model	Accuracy	Key Observation
Logistic Regression	71.7%	Better at identifying employees likely to leave
Random Forest	88.0%	Highest overall prediction accuracy
Gradient Boosting	87.0%	Balanced performance across metrics

Charts & Insights

Employees working overtime showed noticeably higher attrition rates. Lower monthly income employees were more likely to leave the company. Sales Representatives and Laboratory Technicians had higher attrition trends. Employees with poor work-life balance and lower job satisfaction showed higher exit rates. Most resignations occurred during the early years of employment. Younger employees showed slightly higher attrition compared to older employees.



HR Insights & Recommendations

Monitoring employee overtime patterns may help reduce workplace stress and burnout. Additional support for newly joined employees could improve early-stage retention. Regular employee feedback and engagement programs may help identify workplace concerns earlier. Departments with consistently higher attrition trends may benefit from closer employee engagement reviews. Career growth opportunities and skill development initiatives may improve long-term retention. Encouraging healthier work-life balance practices may positively influence employee satisfaction.

Conclusion

This project successfully analyzed employee attrition trends and identified several important workplace factors influencing employee exits. The findings provide meaningful business insights that can support HR teams in improving employee satisfaction, strengthening retention strategies, and making more informed workforce decisions through data-driven analysis.